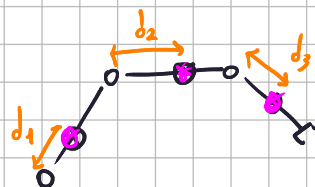
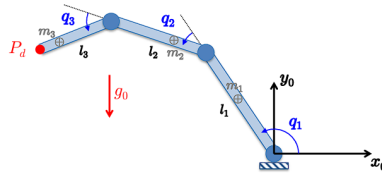


The 3R robot in Fig. 1 moves in a vertical plane. Its control architecture consists of an ideal low-level controller that is able to execute any (reasonable) reference joint velocity command  $\dot{q}_r \in \mathbb{R}^3$  (including  $\dot{q}_r = \mathbf{0}$ ), as received by a high-level control law. The assigned task requires that the end effector is kept at a desired position  $P_d$  in the plane, while the robot minimizes the potential energy  $U_g(q)$  due to gravity. Design a suitable high-level control law that realizes this task in a robust way (i.e., rejecting also positioning errors for the end effector). Provide the detailed symbolic expression of all terms needed in this law. Compute then the numerical value of  $\dot{q}_r$  at the starting instant  $t = 0$ , assuming that the links have a uniform distribution of mass and using the data:

$$l_1 = 0.5, \quad l_2 = 0.4, \quad l_3 = 0.3 \quad [\text{m}], \quad m_1 = 5, \quad m_2 = 3, \quad m_3 = 2 \quad [\text{kg}], \\ q_1(0) = \pi, \quad q_2(0) = 0, \quad q_3(0) = -\pi/2 \quad [\text{rad}] \quad \Rightarrow \quad P_d = (-0.9, 0.3).$$



let  $d_i$  to be the distance between  $RF_{i-1}$  and  $\text{CoM } i$

$$P_{C1x} = d_1 S_1 \quad P_{C2x} = l_1 S_1 + d_2 S_{12} \quad P_{C3x} = l_1 S_1 + l_2 S_{12} + d_3 S_{123} \quad \text{hence}$$

$$U = g_0 \left( m_1 d_1 S_1 + m_2 [l_1 S_1 + d_2 S_{12}] + m_3 [l_1 S_1 + l_2 S_{12} + d_3 S_{123}] \right) \Rightarrow \\ = \underbrace{g_0}_{>0} \left( \underbrace{S_1 [m_1 d_1 + l_1 (m_2 + m_3)]}_{>0} + \underbrace{S_{12} [m_2 d_2 + m_3 l_2]}_{>0} + \underbrace{S_{123} m_3 d_3}_{>0} \right)$$

Observation: minimize  $U$  is equivalent to bring  $S_1, S_{12}$  and  $S_{123}$  to zero, in this priority order. We want to use the projected gradient method:

$$\dot{q}_r = \underbrace{K_p J^\# (P_d - K(q))}_{\substack{\text{go to } P_d \text{ in a} \\ \text{robust way}}} - \underbrace{(\mathbf{I} - J^\# J) \nabla U}_{\substack{\text{projecting } \nabla U \text{ on } \mathcal{NS}(J)}}$$

$$\nabla U = \begin{cases} g_0 (C_1 [m_1 d_1 + l_1 (m_2 + m_3)] + C_{12} [m_2 d_2 + m_3 l_2] + C_{123} m_3 d_3) \\ g_0 (C_{12} [m_2 d_2 + m_3 l_2] + C_{123} m_3 d_3) \\ g_0 C_{123} m_3 d_3 \end{cases}$$

Now consider the numerical values

$$l_{23} = \pi/2$$

$$l_2 = \pi$$

$$l_1 = \pi$$

$$J^\# = \mathbf{B}$$

$$J = \begin{pmatrix} -l_1 S_1 - l_2 S_{12} - l_3 S_{123} & -l_2 S_{12} - l_3 S_{123} & -l_3 S_{123} \\ l_1 C_1 + l_2 C_{12} + l_3 C_{123} & l_2 C_{12} + l_3 C_{123} & l_3 C_{123} \end{pmatrix} = \begin{pmatrix} -0.3 & -0.3 & -0.3 \\ -0.9 & -0.4 & 0 \end{pmatrix} \quad \text{A}$$

$$P_d - K(q) = \begin{pmatrix} 0 & 0 \end{pmatrix}^T \Rightarrow \dot{q}_r = -(\mathbf{I} - J^\# J) \nabla U = (-2.574, 5.792, -3.218)^T$$

$$\mathbf{I} - J^\# J \approx \begin{pmatrix} 0.13 & -0.3 & 0.16 \\ -0.3 & 0.66 & -0.36 \\ 0.16 & -0.36 & 0.204 \end{pmatrix} \quad \nabla U = \begin{cases} -50.5 \\ -13.72 \\ 0 \end{cases}$$

## Exercise #2

Consider again the situation of Exercise #1. Assume now that the robot is torque-controlled, namely that the control architecture is able to impose any (reasonable) reference joint torque command  $\tau_r \in \mathbb{R}^3$ . Design a suitable torque-level control law that realizes the same previous task in a robust way (i.e., rejecting also position and/or velocity errors for the end effector). [Note: You don't have to detail the expression of the terms in the control law, just define the structure.]

We want to impose the following velocity:

$$\dot{q}' = \ddot{J}q + \dot{J}\dot{q}$$

$$\dot{q}_r = K_p J^* (p_d - \kappa(q)) - (I - J^* J) \nabla U \quad \text{consider } \ddot{q}_r$$

$$M\ddot{q} + n = \tau \Rightarrow \ddot{q} = \ddot{M}^{-1}(\tau - n) \quad \text{so we choose}$$

$$\tau = n + M [K_p J^* (p_d - \kappa(q)) - K_d \dot{q} - (I - J^* J) \nabla U] \leftarrow \text{FBL}$$

$$\Rightarrow \ddot{q} = \underbrace{K_p J^* (p_d - \kappa(q))}_{\text{moving towards } p_d} - \underbrace{K_d \dot{q}}_{\text{damping to stop on } p_d} - \underbrace{(I - J^* J) \nabla U}_{\text{moving joints toward argmin } U}$$

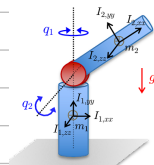
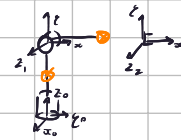
## Exercise #3

Derive the dynamic model of the 2R polar robot in Fig. 2, assuming that  $I_{2,xx} \neq I_{2,yy} = I_{2,zz}$ . Consider also the presence of viscous friction at the two joints.

$$\begin{array}{cccc} \alpha & a & d & \theta \\ \frac{\pi}{2} & 0 & l_1 & q_1 \\ 0 & l_2 & 0 & q_2 \end{array}$$

$${}^0T_1 = \begin{pmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^1T_2 = \begin{pmatrix} c_2 & -s_2 & 0 & l_2 c_2 \\ s_2 & c_2 & 0 & l_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



$${}^1p_{c1} = (0 \quad -d_1 \quad 0)^T \quad {}^2p_{c2} = (-d_2 \quad 0 \quad 0)^T \quad {}^0p_{c1} = (0 \quad l_1 - d_1 \quad 0)^T \Rightarrow U_1 = m g_0 (l_1 - d_1)$$

$${}^1w_1 = {}^0R_1^T (0 \quad 0 \quad \dot{q}_1)^T = (0 \quad \dot{q}_1 \quad 0)^T \quad {}^1v_1 = v_{c1} = 0$$

$${}^2w_2 = {}^1R_2^T (0 \quad \dot{q}_1 \quad \dot{q}_2)^T = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ \dot{q}_1 \\ \dot{q}_2 \end{pmatrix} = (\dot{q}_1 s_2 \quad \dot{q}_1 c_2 \quad \dot{q}_2)^T$$

$${}^2v_2 = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -\dot{q}_1 \dot{q}_1 \\ \dot{q}_2 & 0 \\ -\dot{q}_1 & 0 \end{pmatrix} \begin{pmatrix} l_2 c_2 \\ l_2 s_2 \\ 0 \end{pmatrix} = (0 \quad l_2 \dot{q}_2 \quad -l_2 c_2 \dot{q}_1)^T$$

$$v_{c2} = \begin{cases} 0 \\ \dot{q}_2 (l_2 - d_2) \\ -\dot{q}_1 c_2 (l_2 + d_2) \end{cases} \quad T = \frac{1}{2} I_{1,zz} \dot{q}_1^2 + \frac{1}{2} (I_{2,xx} \dot{q}_1^2 s_2^2 + I_{2,yy} (\dot{q}_1^2 c_2^2 + \dot{q}_2^2)) + \frac{1}{2} m_2 (\dot{q}_2^2 (l_2 - d_2)^2 + \dot{q}_1^2 c_2^2 (l_2 + d_2)^2)$$

$$M_{11} = I_{1,zz} + I_{2,xx} s_2^2 + I_{2,yy} c_2^2 + m_2 c_2^2 (l_2 + d_2)^2 = e_1 + e_2 c_2^2$$

$$M_{22} = I_{2,yy} + m_2 (l_2 - d_2)^2 = e_3$$

$$C_1 = \frac{1}{2} \left( \frac{\partial M_1}{\partial q_1} + \left( \frac{\partial M_1}{\partial q_1} \right)^T - \frac{\partial M}{\partial q_1} \right) = -e_2 c_2 s_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow C_1 = -2e_2 c_2 s_2 \dot{q}_1 \dot{q}_2$$

$$C_2 = \frac{1}{2} \left( \frac{\partial M_2}{\partial q_2} + \left( \frac{\partial M_2}{\partial q_2} \right)^T - \frac{\partial M}{\partial q_2} \right) = e_2 c_2 s_2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow C_2 = e_2 c_2 s_2 \dot{q}_1^2$$

$${}^0P_{C2} = \begin{pmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_2 & -s_2 & 0 & l_2 c_2 \\ s_2 & c_2 & 0 & l_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -d_2 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_2(l_2 - d_2) \\ s_2(l_2 - d_2) \\ 0 \\ 1 \end{pmatrix}$$

$${}^0P_{C2z} = s_2(l_2 - d_2) \Rightarrow U = m_2 g_0 s_2(l_2 - d_2) + \alpha \leftarrow \text{const}$$

$$\Rightarrow g = \begin{pmatrix} 0 & m_2 g_0 c_2(l_2 - d_2) \end{pmatrix}^T = \begin{pmatrix} 0 & g_0 e_4 c_2 \end{pmatrix}^T \quad \text{the model is}$$

$$\begin{cases} (e_1 + e_2 c_2^2) \ddot{q}_1 - 2e_2 c_2 s_2 \dot{q}_1 \dot{q}_2 + b_1 \dot{q}_1 = \tau_1 \\ e_3 \ddot{q}_2 + e_2 c_2 s_2 \dot{q}_1^2 + g_0 e_4 c_2 + b_2 \dot{q}_2 = \tau_2 \end{cases}$$

#### Exercise #4

Consider again the robot of Exercise #3. Answer the following questions on its dynamics.

a) Suppose that the first robot joint is kept at a constant speed  $\dot{q}_1 = \Omega > 0$  by a constant torque  $\bar{\tau}$  of *minimum* possible norm  $\|\bar{\tau}\|$  applied at the joints. Find an associated constant steady-state position  $q_2 = \bar{q}_2$  and the expressions of  $\bar{\tau}_1$  and  $\bar{\tau}_2$ . Is the joint angle  $\bar{q}_2$  unique for a given  $\Omega > 0$ ? And is the associated minimum norm torque  $\bar{\tau}$  unique?

b) Assuming that only the gravity acceleration  $g_0$  and the kinematic parameters are known for this robot, what is the *minimum* number  $p$  of uncertain dynamic coefficients  $\mathbf{a}$  that fully describe the robot dynamics in the linearly parametrized way  $\mathbf{Y}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}) \mathbf{a} = \boldsymbol{\tau}$ ? Give the symbolic expressions for the vector  $\mathbf{a} \in \mathbb{R}^p$  and the  $2 \times p$  matrix  $\mathbf{Y}$ .

$$e_4 = I_{xx} + I_{zz}$$

$$e_5 = I_{zz}$$

$$\text{b)} \quad M_{11} = I_{xx} + I_{zz} s_2^2 + I_{zz} c_2^2 + m_2 c_2^2 (l_2 + d_2)^2 \quad e_1 = I_{xx} + m_2 d_2^2$$

$$M_{22} = I_{zz} + m_2 (l_2 - d_2)^2 = I_{zz} + m_2 l_2^2 + m_2 d_2^2 - 2m_2 l_2 d_2 \quad e_2 = m_2 \quad e_3 = m_2 d_2$$

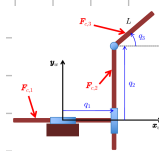
$$M_{11} = e_4 + c_2^2 (e_1 - e_5 + e_2 l_2^2 + 2e_3 l_2) \quad c_1 = -2c_2 s_2 \dot{q}_1 \dot{q}_2 (e_1 - e_5 + e_2 l_2^2 + 2e_3 l_2)$$

$$M_{22} = e_1 + e_2 l_2^2 - 2e_3 l_1 \quad c_2 = \dot{q}_1^2 c_2 s_2 (e_1 - e_5 + e_2 l_2^2 + 2e_3 l_2)$$

$$g_2 = g_0 c_2 e_2 l_2 - g_0 c_2 e_3$$

$$\begin{cases} (e_4 + c_2^2 (e_1 - e_5 + e_2 l_2^2 + 2e_3 l_2)) \ddot{q}_1 - 2c_2 s_2 \dot{q}_1 \dot{q}_2 (e_1 - e_5 + e_2 l_2^2 + 2e_3 l_2) + e_6 \dot{q}_1 = \tau_1 \\ (e_1 + e_2 l_2^2 - 2e_3 l_1) \ddot{q}_2 + \dot{q}_1^2 c_2 s_2 (e_1 - e_5 + e_2 l_2^2 + 2e_3 l_2) + g_0 c_2 e_2 l_2 - g_0 c_2 e_3 + e_7 \dot{q}_2 = \tau_2 \end{cases}$$

A PPR robot moving on a horizontal plane may collide with some unknown obstacle in the environment at an a priori unknown point  $P_c$  along its structure, as illustrated in Fig. 3. Suppose that a single collision occurs and that the interaction can be assumed as pointwise, modeled by an unknown and unmeasurable pure force  $F_{c,i} \in \mathbb{R}^2$  applied to the robot, respectively with  $i = 1, 2$ , or 3 according to which link is involved. Define the complete expression of the dynamic terms in a model-based residual vector  $r \in \mathbb{R}^3$  that can be used for collision detection and isolation. For every possible situation, analyze if the collision can be detected or not, if the colliding link can be isolated or not, if the colliding force  $F_{c,i}$  can be identified (completely or in part) or not, and if the location of the collision point  $P_{c,i}$  along the  $i$ th link can be determined or not.



$$\mathbf{I} \text{ set } \mathbf{r} = \mathbf{K} \left( \mathbf{p} - \int (\mathbf{z} + \mathbf{c} - \mathbf{g} + \mathbf{r}) dt - \mathbf{p}(0) \right)$$

$$\mathbf{v}_{c1} = (\dot{q}_1 \ 0 \ 0) \quad \mathbf{v}_{c2} = (\dot{q}_1 \ \dot{q}_2 \ 0) \quad \mathbf{v}_{c3} = (\dot{q}_1 - d\dot{s}_3 \ \dot{q}_2 + d\dot{c}_3 \ \dot{q}_3)$$

$$\mathbf{T} = \frac{1}{2} M_1 \dot{q}_1^2 + \frac{1}{2} M_2 (\dot{q}_1^2 + \dot{q}_2^2) + \frac{1}{2} \mathbf{I}_3 \dot{q}_3^2 + \frac{1}{2} M_3 (\dot{q}_1^2 + \dot{q}_2^2 + d^2 \dot{q}_3^2 - 2d\dot{s}_3 \dot{q}_1 \dot{q}_3 + 2d\dot{c}_3 \dot{q}_2 \dot{q}_3)$$

$$M_{11} = M_1 + M_2 + M_3 \quad M_{12} = -2d\dot{s}_3 M_3 \quad M_{23} = 2d\dot{c}_3 M_3$$

$$M_{22} = M_2 + M_3 \quad M_{33} = \mathbf{I}_3 + d^2 M_3$$

$$\mathbf{M} = \begin{bmatrix} e_1 & 0 & -2e_2\dot{s}_3 \\ 0 & e_3 & 2e_2\dot{c}_3 \\ -2e_2\dot{s}_3 & 2e_2\dot{c}_3 & e_4 \end{bmatrix} \Rightarrow \mathbf{P} = \begin{cases} e_1 \dot{q}_1 - 2e_2\dot{s}_3 \dot{q}_3 \\ e_3 \dot{q}_2 + 2e_2\dot{c}_3 \dot{q}_3 \\ -2e_2\dot{s}_3 \dot{q}_1 + 2e_2\dot{c}_3 \dot{q}_2 + e_4 \dot{q}_3 \end{cases}$$

$$\mathbf{U} = \mathbf{g}_0 (M_2(q_2 - d_2) + M_3(q_2 + d\dot{s}_3)) \Rightarrow \quad \mathbf{g}_1 = 0 \quad \mathbf{g}_2 = \mathbf{g}_0 e_3 \quad \mathbf{g}_3 = \mathbf{g}_0 e_2 \dot{c}_3$$

$$C_1 = -2e_2\dot{c}_3 \dot{q}_3^2 \quad C_2 = -2e_2\dot{s}_3 \dot{q}_3^2 \quad C_3 = 0$$